Online PES Decoder Learning in Neuromorphic Residual Control of UAV Quaternion Tracking

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***Abstract*— Online neuromorphic residual control can augment a classical attitude controller under abrupt missions or unforeseen disturbances without changing its feedback gains or actuator interface. We present six continuous coupled membrane-gate (CMG) units and eighteen decoder weights adapted by prescribed error sensitivity (PES), adding a bounded correction to the saturated quaternion proportional-derivative (PD) torque request before rotor-force allocation. Across ten matched 45-second MuJoCo cases on the Skydio X2 and a Crazyflie 2 derived model, mean quaternion-angle error decreases by 17.3–60.6%, including approximately 53–57% reductions on the Full3 and Hard45 multi-axis tracking missions. Direct and Nengo implementations agree closely. Benchmarks against LIF, ALIF, Izhikevich, and LIFRate neuron models expose an effort-accuracy tradeoff: CMG–PES attains better tracking with substantially lower realized residual effort. Under synthetic measurement noise, the error reductions persist, and learned corrections are reused across repeated disturbances. Overall, PD provides a nominal, interpretable reference, while the online residual adds bounded, revocable adaptation that improves robustness to disturbances and noisy measurements.**

I. Introduction

Learning-based augmentation can retain classical feedback while adapting a correction that is difficult to specify *a priori*. We study a bounded torque residual for quaternion attitude tracking of an unmanned aerial vehicle (UAV) with individual rotor-force saturation. The question is whether a small stateful basis improves a fixed PD controller, and whether any improvement comes from its neuron dynamics, decoder learning, or correction magnitude. Bounded authority makes the learning layer switchable; it does not by itself ensure stability or safety.

Prior work includes residual robot learning [1], geometric $\mathcal{L}_1$ quadrotor augmentation with performance guarantees [2], and proportional–integral–derivative (PID) attitude control augmented with a proximal policy optimization (PPO) residual [3]. Neuromorphic control offers a complementary implementation approach: Nengo supports the Neural Engineering Framework (NEF) and online decoder learning [4], most often through the prescribed error sensitivity (PES) rule, an established local, online error-driven update [5]. PES-based adaptive control has been demonstrated on manipulators [6], [7], and in a Nengo spiking PID controller for a free-flying UAV [8]. Building on that line, our focus is a continuous membrane-gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

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We augment a quaternion PD baseline with a compact neuromorphic residual architecture based on coupled conductance-based membrane-gate (CMG) units and PES decoder learning. The paper's contributions are:

1) a code-aligned CMG–PES architecture with paired currents, explicit encoding, state-update, and decoder/readout equations, and a fully specified actuator interface (Fig. 1);
2) proofs of internal-state, command-deviation, and weight boundedness, with exact recovery of the PD baseline at zero residual authority;
3) a controlled ablation that separates conductance modulation, decoder learning, and output authority, and identifies which mechanism is primarily responsible for the tracking-error reduction; and
4) robustness evidence that the reduction survives realistic measurement noise, together with a hold-equilibrium study in which the learned correction is reused across repeated disturbance episodes rather than relearned.

The CMG unit couples a leaky membrane to a gating state that modulates its effective conductance, linking LIF-like membrane integration with Hodgkin–Huxley inspired gating dynamics. Although related operating regimes can support spike-like responses, the present controller deliberately exploits the continuous, non-spiking membrane-gate trajectory, which provides a compact set of stateful temporal features for the residual. The present implementation is digital; the same temporal-feature mechanism holds regardless of computing platform, and no neuromorphic-hardware energy advantage is claimed.

The remainder of the paper is organized as follows. Section II defines the attitude-tracking problem, notation, and actuator interface. Section III presents the CMG–PES residual architecture, state dynamics, and learning rule. Section III establishes boundedness properties. Section IV reports the simulation study, ablations, and benchmarks, including noise and repeated-disturbance experiments. Section V concludes and discusses limitations and future work.

II. Attitude Baseline and Actuator Interface

The objective is to reduce shortest-path attitude error for a prescribed $q_d(t)$ while respecting rotor-force limits. Baseline and augmented controllers share the state measurements, height loop, and allocator; Section III modifies only the requested attitude torque.

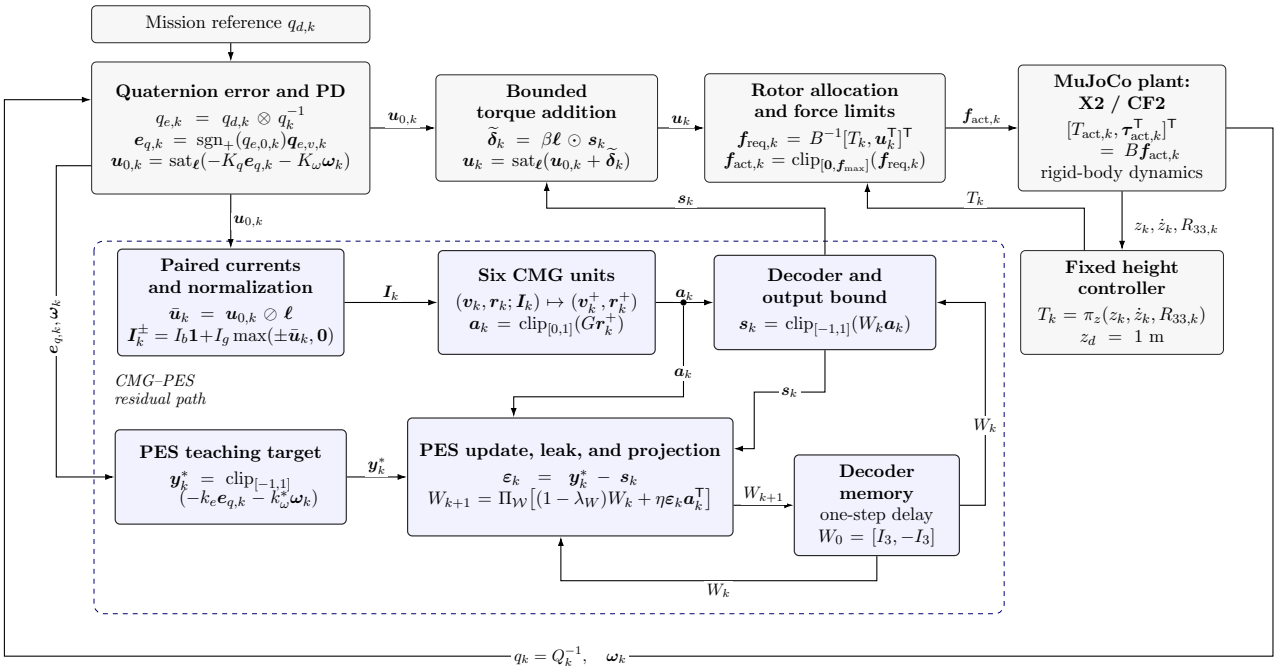


Fig. 1. Online PES decoder learning CMG-PES Neuromorphic Residual Control architecture



Fig. 2. Vehicles and simulation models. Left: CF2 hardware and MuJoCo model. Centre: X2 hardware and MuJoCo model. Right: both geometries at relative scale. The axis triads are body-frame axes: they are rigidly attached to the vehicle and therefore rotate with it relative to the inertial (world) frame (cf. Q: body $\to$ inertial in Sec. II).

A. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ denote the scalar-first Hamilton quaternion representing attitude (body to inertial), and let $\boldsymbol{\omega}$ be the body angular velocity. For a rigid body with inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes (cf. Fig. 2), the free-flight rotational dynamics are [9]

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\mathrm{p}}$ denotes modeled passive torque and $\boldsymbol{\tau}_{\mathrm{d}}$ any applied disturbance; the nominal ideal rigid-body model sets both to zero. Contact, if present, contributes an additional external wrench.

We work with the corresponding passive quaternion $q = Q^{-1}$ [10], which satisfies

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T} \otimes q. \tag{3}$$

The desired attitude q_d uses the same convention. Define the attitude error

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{4}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0}) \boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\,\mathrm{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The reported error θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$. Sign selection introduces a discontinuity at $q_{e,0} = 0$; it does not establish robust hybrid global stability [11].

For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^\mathsf{T}]^\mathsf{T}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^\mathsf{T} \boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2} \boldsymbol{q}_{e,v}^\mathsf{T} \boldsymbol{\omega}_e. \tag{8}$$

These are kinematic identities (no convergence implied). Reference Euler triples parameterize q_d in this passive convention.

B. Physical Inertia and Implemented PD Gains

Let $\boldsymbol{J} \succ 0$ denote the rigid-body inertia about the center of mass, expressed in body axes [9]. When principal moments $\boldsymbol{j}_{\mathrm{p}}$ are specified in a rotated principal-axis frame, the physical body-frame tensor is

$$\boldsymbol{J} = R_{BI}\,\mathrm{diag}(\boldsymbol{j}_{\mathrm{p}})R_{BI}^{\mathsf{T}}, \tag{9}$$

for the rotation from principal axes to body axes R_{BI}. The implementation sets axiswise gains using the principal moments in their reported ordering, $\boldsymbol{j}_{\mathrm{tune}}$, not necessarily $\mathrm{diag}(\boldsymbol{J})$ (Table I).

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta \boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n, \tag{10}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{11}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^{\mathsf{T}}$ rad/s, $\zeta = 0.95$, and $\mathrm{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. The requested-torque box is $\mathcal{U} = \prod_{i=1}^{3}[-\ell_i, \ell_i]$; $\mathrm{clip}_{[a,b]}$ also acts componentwise, and $\oslash$ denotes componentwise division. We impose componentwise torque limits $\boldsymbol{\ell} \in \mathbb{R}^3_{>0}$ through $\mathrm{sat}_{\boldsymbol{\ell}}(\cdot)$. In our implementation, $\boldsymbol{\ell}$ is chosen proportional to the diagonal gain-scheduling vector $\boldsymbol{j}_{\mathrm{tune}}$ via $\ell_i = \max(j_{\mathrm{tune},i}\nu_i, 10^{-6}\ \mathrm{N\,m})$, with acceleration-scale vector $\boldsymbol{\nu} = (12, 12, 10)^{\mathsf{T}}$ rad/s^2. These controller-side bounds are not the rotor-feasible torque set. The law is quaternion feedback [12]; $\boldsymbol{\omega}_n$ is a tuning parameter rather than a claimed set of coupled closed-loop natural frequencies. Desired-rate feedforward and gyroscopic cancellation are omitted. The resulting gains are vehicle-specific through $\boldsymbol{j}_{\mathrm{tune}}$, even though $\boldsymbol{\omega}_n$ and ζ are shared; the same fixed gains are then retained for PD and its CMG augmentation. The local result in Section III-D uses these anisotropic gains; it does not replace them with a scalar-gain controller.

C. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{12}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^{\mathsf{T}}$ and the actuator's local reaction torque $\kappa_i f_i$. Here B maps rotor thrusts to total thrust and body torque; κ_i has units of meters, not angular velocity. Given a commanded collective thrust T and body-torque command $\boldsymbol{u}$, the corresponding inverse-allocation and clipping model is

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{\dagger}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}}, \\ \boldsymbol{f}_{\mathrm{act}} &= \mathrm{clip}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^{\mathsf{T}}]^{\mathsf{T}} &= B\boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{13}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{\dagger}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}} \in [\boldsymbol{0}, \boldsymbol{f}_{\max}]$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [13].

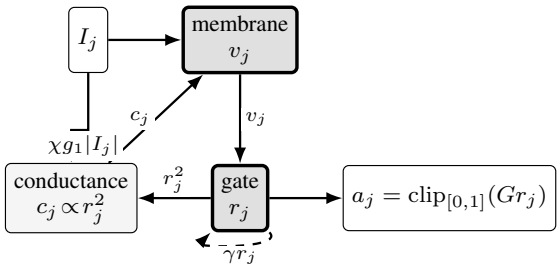


Fig. 3. Coupling topology of one CMG unit, cf. (18). The membrane sets the gate target; the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$ ($\chi = 1$ makes that shunt input dependent). The dynamics are continuous: no spike, no reset, and only a_j leaves the unit.

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0,2.5Mg]}\left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)}\right). \tag{14}$$

Here g is the gravitational acceleration. The height loop reads altitude and vertical velocity; the denominator compensates tilt. The guard 0.35 prevents near-singular tilt compensation as $R(Q)_{33} \to 0$ (i.e., near 90° tilt) and corresponds to limiting the effective compensation to about $\arccos(0.35) \approx 69.5^\circ$. Its upper bound does not supersede individual rotor limits. The CMG path has no direct collective output, but can affect height through attitude, tilt compensation, and allocation saturation. The resulting shared plant interface isolates changes in requested attitude torque while preserving the same physical simulation for both controllers.

III. CMG State Dynamics and Online Decoder Adaptation

This section specifies the residual architecture in state-space form and the online decoder update, then states the boundedness properties that follow from the resulting saturations and projections.

A. Encoding, States, and Numerical Update

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^{\pm} &= I_b\boldsymbol{1} + I_g \max(\pm\bar{\boldsymbol{u}}, \boldsymbol{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^{+})^{\mathsf{T}}, (\boldsymbol{I}^{-})^{\mathsf{T}}]^{\mathsf{T}}. \end{aligned} \tag{15}$$

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{16}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{17}$$

$$c_j = (g_0 + \chi g_1 |I_j|)r_j^2, \qquad \chi \in \{0, 1\}. \tag{18}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 3). Thus,

both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\text{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\text{th}} + \gamma r_{j,k})]\Big), \tag{19}$$

$$r^+_{j,k} = e^{-h/\tau_r} r_{j,k} + (1 - e^{-h/\tau_r}) b_{j,k}. \tag{20}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1 |I_{j,k}|)(r^+_{j,k})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\text{rest}} + I_{j,k})/\tau + c_{j,k} v_{\text{reset}}$. The frozen-coefficient membrane step is

$$v^+_{j,k} = e^{-\lambda_{j,k} h} v_{j,k} + (1 - e^{-\lambda_{j,k} h}) d_{j,k}/\lambda_{j,k}. \tag{21}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v^+_{j,k}, r^+_{j,k})$.

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \text{clip}_{[0,1]}(G \boldsymbol{r}^+_k), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \text{clip}_{[-1,1]}(W_k \boldsymbol{a}_k). \end{aligned} \tag{22}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. With $G = 10^5$ the activity saturates once $r_j \geq 10^{-5}$, so G is a primary design hyperparameter for how much of r_j's dynamic range reaches the decoder. The teaching target and local update are

$$\boldsymbol{y}^*_k = \text{clip}_{[-1,1]}(-k_e \boldsymbol{e}_{q,k} - k^*_\omega \boldsymbol{\omega}_k), \tag{23}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}^*_k - \boldsymbol{s}_k, \tag{24}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta \boldsymbol{\varepsilon}_k \boldsymbol{a}_k^\mathsf{T} - \lambda_W W_k]. \tag{25}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m \times 2m}$. The coefficients η and λ_W are per-step (so changing h changes the effective learning dynamics). PES is a local error-driven update toward the prescribed target $\boldsymbol{y}^*_k$; it does not differentiate through the plant or claim to minimize a closed-loop cost. Furthermore, because the network outputs a normalized fractional command $\boldsymbol{s}_k$, the PES target safely relies on the downstream baseline authority limits ($\boldsymbol{\ell}$) for its physical anisotropic scaling, avoiding the need for axis-specific teaching gains.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \text{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta \boldsymbol{\ell} \odot \boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \quad 0 \leq \beta \leq 1. \tag{26}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

We first state the boundedness properties that follow directly from the architecture: internal-state bounds, a uniform command-deviation bound set by residual authority, and projection-based bounds on decoder weights.

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table II, $r_{j,0} \in [0,1]$, and for any $h > 0$, the gate remains in $[0,1]$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\text{rest}} + I_b, v_{\text{rest}} + I_b + I_g]$.*

Proof. Because $b_{j,k} \in [0,1]$, the exact exponential Euler gate update is a strict convex combination, unconditionally guaranteeing $r \in [0,1]$ for any step size $h > 0$ without requiring a clip operator. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \geq 0$, d_j/λ_j is a convex combination of $v_{\text{rest}} + I_j$ and v_{reset}. Equation (21) is a convex combination of $v_{j,k}$ and d_j/λ_j. Induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \leq \beta \ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness and $\text{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \leq |\beta \ell_i s_{k,i}| \leq \beta \ell_i$. The recovery statements follow by substitution. □

The same allocator and state then give the same actuator output when $\beta = 0$. The proposition does not establish state safety, rotor headroom, stability, or a bound on tracking improvement. A bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then (25) keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta \|\boldsymbol{a}\|_2^2) \boldsymbol{\varepsilon}_k. \tag{27}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta \boldsymbol{\varepsilon}_k \|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (27). The scalar multiplier has modulus less than one under the stated condition. Furthermore, the readout clip $\text{clip}_{[-1,1]}$ and weight projection $\Pi_{\mathcal{W}}$ are non-expansive projections onto closed convex sets. Since the target $\boldsymbol{y}^*_k$ resides strictly within $[-1,1]^m$, applying the clip mathematically guarantees $\|\text{clip}(\boldsymbol{z}_k) - \boldsymbol{y}^*_k\| \leq \|\boldsymbol{z}_k - \boldsymbol{y}^*_k\|$, preventing saturation-induced parameter divergence. □

This is the constant-feature PES result [5], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \leq 2m$, $0 < \eta < 1/m$ is sufficient for that restricted model with nonzero activity; changing targets, loss of excitation, clipping, and leakage

invalidate a direct application to the experiment. No unique identification of all decoder weights follows from one fixed regressor.

D. Allocation-Aware Local Stability

Let $E = [\mathbf{0}_{3\times1}, I_3]^\mathsf{T}$, $H = E^\mathsf{T}$, and let $\mathcal{A}_T(\boldsymbol{u}) = HB\,\mathrm{clip}_{[0,\boldsymbol{f}_{\max}]}(B^\dagger[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T})$ be the realized torque at collective T. For two commands at the *same state and collective*, scalar clipping has secant slopes $d_i \in [0,1]$. Consequently,

$$\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0) = HBDB^\dagger E\boldsymbol{\delta}, \tag{28}$$

$$\|\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0)\|_2 \le L_A\beta\|\boldsymbol{\ell}\|_2, \tag{29}$$

where $D = \mathrm{diag}(d_i)$ and $L_A = \max_{D\in\mathcal{D}} \|HBDB^\dagger E\|_2$. $\mathcal{D}$ is the set of sixteen diagonal matrices with entries in $\{0,1\}$. The vertex maximum bounds all intermediate D by convexity. Note that the unweighted 2-norm L_A is artificially inflated by weak yaw authority ($|\kappa_i| \ll |p_{i,x}|, |p_{i,y}|$). Utilizing a diagonal weighting matrix $W = \mathrm{diag}(1, 1, c_\psi)$ to normalize authority before computing the spectral norm would yield a tighter, physically realistic Lipschitz constant $L_{A,w}$. This holds even with rotor clipping, but is not a bound between two different closed-loop trajectories. It also does not bound the height error.

For a constant reference and a smooth shortest-path branch, let $\boldsymbol{x} = [\boldsymbol{e}_q^\mathsf{T}, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$ and $K_q = \mathrm{diag}(\boldsymbol{k}_q)$, $K_\omega = \mathrm{diag}(\boldsymbol{k}_\omega)$. Write the continuous-time attitude subsystem as

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x}) + F\boldsymbol{d},$$
$$A = \begin{bmatrix} 0 & \frac{1}{2}I_3 \\ -\boldsymbol{J}^{-1}K_q & -\boldsymbol{J}^{-1}K_\omega \end{bmatrix}, \qquad F = \begin{bmatrix} 0 \\ \boldsymbol{J}^{-1} \end{bmatrix}, \tag{30}$$

where $\boldsymbol{d} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} + K_q\boldsymbol{e}_q + K_\omega\boldsymbol{\omega}$ is the total deviation from ideal PD torque. It includes allocation and command-saturation errors, not just the neural request. The nonlinear remainder is

$$\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} \frac{1}{2}[(\sqrt{1-\|\boldsymbol{e}_q\|^2} - 1)I_3 + [\boldsymbol{e}_q]_\times]\boldsymbol{\omega} \\ -\boldsymbol{J}^{-1}[\boldsymbol{\omega}]_\times\boldsymbol{J}\boldsymbol{\omega} \end{bmatrix} = O(\|\boldsymbol{x}\|^2), \tag{31}$$

with $[\boldsymbol{z}]_\times\boldsymbol{y} = \boldsymbol{z}\times\boldsymbol{y}$. Norms below use fixed coordinates with angular rates expressed in rad/s. Because the scalar kinematic term $(\sqrt{1-\|\boldsymbol{e}_q\|^2} - 1)\boldsymbol{\omega}$ is orthogonal to the cross-product $[\boldsymbol{e}_q]_\times\boldsymbol{\omega}$, their squared norms are additive. Using the algebraic inequality $(1 - \sqrt{1-\|\boldsymbol{e}_q\|^2})^2 \le \|\boldsymbol{e}_q\|^2$ and the AM-GM inequality, the nonlinear remainder possesses the exact closed-form Lipschitz constant $L_r = \frac{\sqrt{2}}{4} + \kappa(\boldsymbol{J})$, where $\kappa(\boldsymbol{J})$ is the condition number of the physical inertia tensor.

Theorem 1 (Conditional local practical stability)**.** *Let $\boldsymbol{J}, K_q, K_\omega \succ 0$, let q_d be constant, and use the smooth branch in* (30)*. Choose $Q_L \succ 0$ and the solution $P \succ 0$ of $A^\mathsf{T}P + PA = -Q_L$. On $\|\boldsymbol{x}\| \le r < 1$, suppose $\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2$ using the exact closed-form Lipschitz bound $L_r = \frac{\sqrt{2}}{4} + \kappa(\boldsymbol{J})$, and choose r so that $c = \lambda_{\min}(Q_L) - 2\|P\|L_r r > 0$. Set $p_- = \lambda_{\min}(P)$, $p_+ = \lambda_{\max}(P)$, $b = \|PF\|$, and $V = \boldsymbol{x}^\mathsf{T}P\boldsymbol{x}$. If*

$$V(0) \le p_- r^2, \qquad \|\boldsymbol{d}(t)\| \le \bar{d} < \frac{cr}{2b}\sqrt{\frac{p_-}{p_+}}, \tag{32}$$

then $V \le p_- r^2$ is forward invariant and

$$V(t) \le e^{-\alpha t}V(0) + \frac{4p_+b^2}{c^2}\bar{d}^2, \qquad \alpha = \frac{c}{2p_+}. \tag{33}$$

Thus attitude regulation is locally input-to-state stable; zero perturbation gives local exponential convergence of the plant state.

Proof. Define the linear mechanical energy $V_L = \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{e}_q + \frac{1}{2}\boldsymbol{\omega}^\mathsf{T}\boldsymbol{J}\boldsymbol{\omega}$. Its time derivative along the linear dynamics is $\dot{V}_L = \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{\omega} - \boldsymbol{\omega}^\mathsf{T}K_q\boldsymbol{e}_q - \boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega}$. Because the implemented gain $K_q = \mathrm{diag}(\boldsymbol{k}_q)$ is strictly symmetric, the stiffness cross-terms perfectly cancel regardless of the full physical inertia tensor $\boldsymbol{J}$'s asymmetry, yielding $\dot{V}_L = -\boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega} \le 0$. By LaSalle's Invariance Principle, on the invariant set where $\dot{V}_L \equiv 0$, we have $\boldsymbol{\omega} \equiv \mathbf{0} \implies \dot{\boldsymbol{\omega}} \equiv \mathbf{0}$, which from the unforced linear momentum equation requires $\boldsymbol{e}_q = \mathbf{0}$. The origin is the largest invariant set, making A unconditionally Hurwitz and ensuring $P \succ 0$ uniquely exists. Equations (30)–(31) give

$$\dot{V} \le -c\|\boldsymbol{x}\|^2 + 2b\|\boldsymbol{x}\|\bar{d} \le -\frac{c}{2}\|\boldsymbol{x}\|^2 + \frac{2b^2}{c}\bar{d}^2.$$

On $V = p_-r^2$, $\|\boldsymbol{x}\| \ge r\sqrt{p_-/p_+}$, so (32) makes the first bound strictly negative. The ellipsoid stays inside the radius-r ball; integration of $\dot{V} \le -\alpha V + 2b^2\bar{d}^2/c$ gives (33). □

If the same-state baseline deviation from ideal PD is bounded by $\bar{d}_0$, then (29) permits $\bar{d} = \bar{d}_0 + L_A\beta\|\boldsymbol{\ell}\|$ in the theorem, *provided* (32) holds. This is a sufficient local robustness condition, independent of the particular PES trajectory, not a PES convergence proof. Moving references, the 180° sign boundary, sampled-data/solver error, and full translational stability remain outside this result; sample-and-hold error would need its own bound within $\boldsymbol{d}$. By optimizing the invariant set radius to $r^* = \frac{\lambda_{\min}(Q_L)}{4\|P\|L_r}$, we obtain the Ultimate Robustness Margin: $\bar{d}_{\max} = \frac{\lambda^2_{\min}(Q_L)}{16b\|P\|L_r}\sqrt{\frac{p_-}{p_+}}$. Equating this optimal margin to the bounded allocation deviation yields an explicit design law for the maximum mathematically safe neural authority: $\beta_{\max} = \frac{\bar{d}_{\max} - \bar{d}_0}{L_A\|\boldsymbol{\ell}\|_2}$.

Remark 1 (Topological Safety)**.** *Because the invariant bounding ellipsoid strictly enforces $\|\boldsymbol{x}\| \le r < 1$, it mathematically guarantees $\|\boldsymbol{e}_q\| < 1$. Consequently, $q^2_{e,0} = 1 - \|\boldsymbol{e}_q\|^2 > 0$. The system is dynamically trapped in the positive hemisphere, guaranteeing the $q_{e,0} = 0$ switching manifold is never reached, thus robustly avoiding unwinding.*

a) Nengo realization.: In Nengo 4.1.0, a custom `NeuronType` exposes membrane and gate states. Direct current injection into `ensemble.neurons` bypasses NEF gain/intercept fitting. Unfiltered CMG activity uses $\eta_N = \eta(2m)/h$ and Nengo error $\boldsymbol{s} - \boldsymbol{y}^*$. Leak uses pre-step weights; projection follows the simulator step. Nengo's teaching graph introduces a one-step latency, so this is not an identical weight sequence. The maximum relative mean-error difference over ten direct/Nengo cases is 2.013×10^{-6}, a tested numerical agreement rather than formal backend equivalence.

TABLE I

MAIN MODEL/INTERFACE PARAMETERS AND MANUFACTURER CONTEXT. MASS AGREES WITH PUBLISHED CF2/X2 SPECIFICATIONS [17], [18]; FORCE LIMITS, INERTIAS, AND REACTION COEFFICIENTS ARE SIMULATION PARAMETERS, NOT IDENTIFIED HARDWARE RATINGS.

Parameter	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction magnitude $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Rotor force limit (N)	$.0875 \to .13125$	13
Total force limit (N)	$.350 \to .525$	52
Main-test thrust/weight	1.982	4.001
$\boldsymbol{j}_{\mathrm{tune}}$ (10^{-5} kg m^2)	$(2.3951, 2.3951, 3.2347)$	—
$\boldsymbol{j}_{\mathrm{tune}}$ (10^{-2} kg m^2)	—	$(6.0711, 3.6468, 2.5412)$
Rigid-body integrator	RK4	Euler
Experiment timestep (s)	0.0005	0.0005
Manufacturer thrust data	.568 N[a]	Not published[b]

[a]Historical CF2 measurement at 93.75% PWM using an external power supply [19]; not a continuous battery-powered rating. The modeled .525 N is below that observation. [b]Not reported in the reviewed X2 datasheet. Neither model includes motor lag; a historical CF2 motor step took about 180 ms [20], illustrating an omitted hardware effect.

IV. EXPERIMENTAL PROTOCOL AND CROSS-MISSION RESULTS

A. Models, Parameters, and Measurements

MuJoCo provides rigid-body simulation with site-applied actuator wrenches [14]. We evaluate two Menagerie-derived multirotor models [15] (CF2 and X2) as free-flying rigid bodies: the attitude loop commands three body torques and a separate height loop commands collective thrust.

The upstream CF2 model exposes abstract moment channels [16]; we replace them with four rotor-force channels and the shared clipped inverse allocator (13) to expose thrust/torque coupling and per-rotor limits. Rotor ordering is $(-x,-y),(-x,+y),(+x,+y),(+x,-y)$ with reaction signs $(-,+,-,+)$. Table I separates simulation/interface parameters from manufacturer context; the CF2 $1.5\times$ scale increases a model limit, not measured hardware capability. Both models are simplified (no motor/RPM/battery states and no validated rotor aerodynamics); feedback uses `qpos`/`qvel`, with noise injected only in Sec. IV-C.

CF2's body axes are principal axes. For X2, the geometry-derived body tensor is

$$\boldsymbol{J} = 10^{-2}\begin{bmatrix} 3.66517 & 0 & -.21000 \\ 0 & 2.54117 & 0 \\ -.21000 & 0 & 6.05280 \end{bmatrix}\ \mathrm{kg\,m^2}.$$

It differs from both $\mathrm{diag}(\boldsymbol{j}_{\mathrm{tune}})$ and a manufacturer-identified tensor. Torque limits use $\boldsymbol{j}_{\mathrm{tune}}$ as defined in Section II.

Each PD/CMG pair uses the same initial state, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic (no injected disturbances or sensor noise), with one run per cell and learning active throughout. We report the mean error $\bar{\theta}_q$ and relative reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$; means use all 90,000 samples even when plotted traces are decimated.

TABLE II

CANONICAL 3-CHANNEL CMG–PES PARAMETERS. NEURAL STATES/CURRENTS ARE NORMALIZED MODEL VARIABLES; LEARNING COEFFICIENTS ARE PER STEP.

Quantity	Value
h, duration	0.0005 s, 45 s
τ, τ_r	0.05 s, 0.005 s
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$
$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
I_b, I_g, G	$0.75, 1.25, 10^5$
$\beta, \eta, \lambda_W, W_{\max}$	$0.5, 8\times10^{-4}, 10^{-6}, 6$
k_e, k_ω^*	$8, 0$

The five reference triples are generated as follows (radians): Roll is $(0.25\sin 0.7t, 0, 0)$; Full3 is $0.5\sin(t + (0,\pi/2,\pi)^{\mathsf{T}})$, componentwise. Hard45 is

$$\begin{bmatrix} 0.55\sin(0.75t) + 0.20\sin(2.10t + 0.40) \\ 0.45\sin(0.95t + \pi/3) + 0.18\sin(1.75t) \\ 0.65\sin(0.55t + \pi) + 0.25\sin(1.35t + 0.20) \end{bmatrix}. \tag{34}$$

Hardstep holds zero for one second, then cycles every two seconds through the rows of

$$\begin{bmatrix} .75 & -.45 & .55 \\ -.65 & .55 & -.70 \\ .50 & .50 & .90 \\ -.80 & -.35 & .20 \end{bmatrix}. \tag{35}$$

Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. The nominal Hover equilibrium is also available, but its essentially zero error does not support a meaningful percentage comparison.

B. Primary Five-Mission Matrix

CMG-PES improves mean quaternion-angle tracking error in all ten nominal missions while respecting the same actuator interface as the PD baseline. The largest reductions occur on Yaw360 and Full3 and the smallest on Roll. Hardstep has the largest absolute mean error and nonzero rotor-bound occupancy on both models. The statistic S measures

TABLE III

CANONICAL NENGO CMG–PES VERSUS PD: MEAN QUATERNION-ANGLE ERROR [RAD], 45 S, $n = 1$ PER CELL. S IS THE LOGGED PERCENTAGE OF SAMPLES WITH AT LEAST ONE ROTOR COMMAND AT A BOUND, PD/CMG. CF2 DENOTES THE $1.5\times$ ROTOR-AUTHORITY DESIGN, NOT STOCK HARDWARE.

Model	Mission	PD	CMG–PES	Reduction	S [%]
X2	Roll	.050565	.041785	17.36%	0/0
	Full3	.303275	.131748	56.56%	0/0
	Hard45	.294082	.138007	53.07%	0/0
	Hardstep	.454528	.359355	20.94%	.016/.239
	Yaw360	.075863	.029902	60.58%	0/0
CF2	Roll	.051337	.042439	17.33%	0/0
	Full3	.288383	.126476	56.14%	0/0
	Hard45	.285449	.132319	53.65%	0/0
	Hardstep	.426182	.347886	18.37%	.137/.264
	Yaw360	.075812	.029870	60.60%	0/0

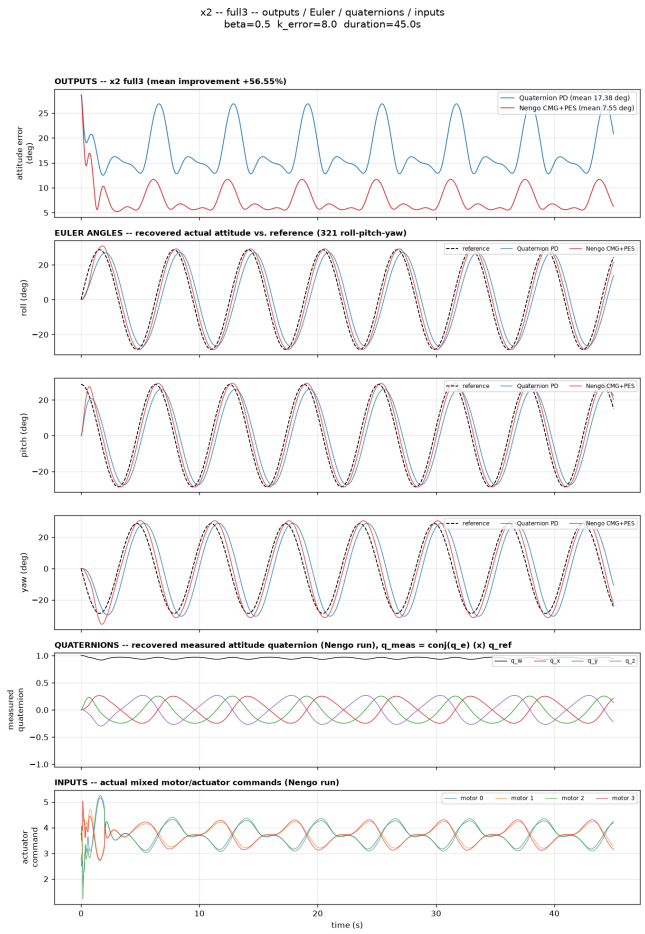


Fig. 4. X2 Full3 mission: PD baseline versus CMG–PES augmentation over the 45-s run. Full3 commands all three axes simultaneously and is the largest reduction that is also insensitive to measurement noise, losing under one point between the clean and realistic columns of Table IV.

post-clipping commands near a limit, not requested-wrench infeasibility magnitude. The results establish deterministic nominal performance relative to this fixed PD law, not optimality over classical controllers.

Figures 4 and 5 pair tracking performance and internal CMG signals for X2 Full3. They contrast smooth multi-axis tracking with the corresponding neuromorphic internal dynamics. Improvement refers to the mean metric; it need not hold at every instant.

C. Sensitivity to Measurement Noise

Each of the ten cases was rerun at two synthetic noise levels and two paired seeds. At every step, independent zero-mean Gaussian samples form a small Euler-angle quaternion left-multiplied into q, and additive Gaussian samples corrupt each angular-rate component. These levels are not calibrated sensor specifications. Crucially, the runner evaluates θ_q from the *corrupted* quaternion used by the controller, not from an independent ground-truth attitude. Table IV therefore measures noisy-feedback error.

All reported noisy-feedback reductions remain positive. At the higher level they range from 18.08 to 55.75%; Yaw360 loses about 12.8 pp relative to clean feedback, whereas Roll's relative reduction increases. Sample SD ranges are .001–.101 pp at low noise and .002–.031 pp at high noise. Two seeds demonstrate limited repeatability, not a reliable population confidence interval or effectively deterministic behavior.

TABLE IV

MEAN-ERROR REDUCTION VERSUS PD UNDER MEASUREMENT NOISE. ENTRIES ARE X2/CF2. CLEAN MATCHES THE NOISE-FREE PRIMARY-MATRIX RESULTS; MODERATE IS 0.010 RAD AND 0.025 RAD S^{-1}, REALISTIC IS 0.020 RAD AND 0.050 RAD S^{-1} ON THE ATTITUDE AND RATE ESTIMATES. NOISY ENTRIES AVERAGE TWO SEEDS; THE SEED STANDARD DEVIATION IS GIVEN FOR THE REALISTIC LEVEL.

Mission	Clean	Moderate	Realistic
Roll	17.4/17.3%	32.3/32.4%	32.2/32.5% ±0.01
Full3	56.6/56.1%	56.4/55.9%	55.7/55.2% ±0.00
Hard45	53.1/53.6%	52.9/53.4%	52.3/52.7% ±0.01
Hardstep	20.9/18.4%	20.8/18.3%	20.6/18.1% ±0.02
Yaw360	60.6/60.6%	56.8/56.8%	47.8/47.8% ±0.03

D. Ablation: What Drives the Improvement?

We separate decoder adaptation (PES), input-dependent conductance modulation (χ), and readout gain (G). Toggling χ, which retains the base shunt $g_0 r_j^2$, is numerically indistinguishable with the decoder mode held fixed. *Fixed versus adaptive decoder.* Holding $W_k = W_0$ keeps the neuron dynamics but reduces the readout to $\boldsymbol{a}^+ - \boldsymbol{a}^-$, a bounded, stateful image of the PD request. It still helps (X2 0.12–10.27%, CF2 0.06–6.80%; Table III), so fixing the decoder does not remove *all* improvement, yet it recovers at most a fifth of the PES gain in any cell. Fig. 5 shows why, row by row. *Membrane and gate.* At $I_j = I_b$ the membrane rests at $v_j \approx 0.75 < v_{\mathrm{th}}$, so $r_j = \sigma(\alpha_h(I_b - v_{\mathrm{th}})) = \sigma(-12.5) \approx 3.7 \times 10^{-6}$: the gate is deep in its logistic tail, not closed. Once $r_j \geq 1/2$, $\gamma = 2$ and $v_j \geq 0$ (Proposition 1) give $b_{j,k} \geq 1/2$ in (20), so an opened gate cannot close (none did in the twenty runs), and the shunt holds v_j near 0.09. *Activity.* With $G = 10^5$ the resting gate already gives $a_j \approx 0.37$, graded by the input; a latched gate pins $a_j = 1$. With the fixed decoder only the $\pm y$ pair latches on Full3 (both models), so $a_y^+ - a_y^- = 0$ and the pitch residual is exactly zero after 1 s while the sub-threshold units carry roll and yaw. Under PES on X2, $\boldsymbol{a}_k \equiv \mathbf{1}$ from 2.02 s (all gates latched by 2.4 s), and for constant $\bar{\boldsymbol{a}}$ with inactive projection the direct update gives $\boldsymbol{z}_k = W_k \bar{\boldsymbol{a}}$ the recursion

$$\boldsymbol{z}_{k+1} = (1 - \lambda_W)\boldsymbol{z}_k + \eta \|\bar{\boldsymbol{a}}\|^2 [\boldsymbol{y}_k^* - \mathrm{clip}_{[-1,1]}(\boldsymbol{z}_k)], \quad (36)$$

the leaky constant-feature regime of Proposition 3: a first-order filter of the teaching target with pole 0.9952 (about 0.10 s) while the readout clip is inactive. Because the latched CMG gates ($\boldsymbol{a}_k \approx \mathbf{1}$) collapse the discrete-time PES update into a first-order low-pass filter (Leaky Integrator) acting purely on proportional position error, the neural network mathematically functions as a classical Adaptive Lag Compensator (Integral Action). This formally explains its ability to erase steady-state errors without amplifying high-frequency measurement noise. *Residual and decoder.* The

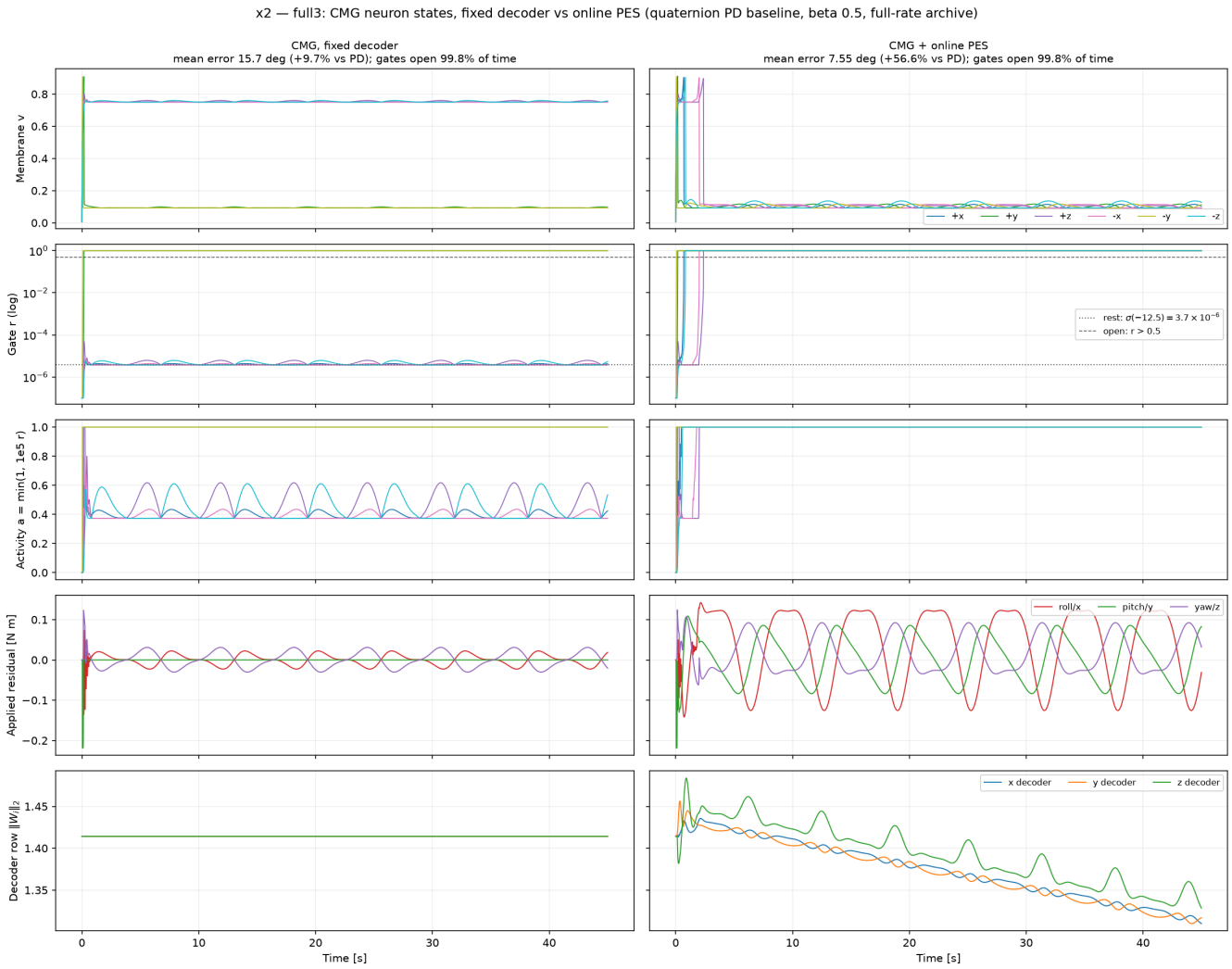


Fig. 5. X2 Full3: internal CMG signals on the trajectory of Fig. 4, comparing fixed-decoder CMG (left) to online PES adaptation (right). Rows separate membrane state, raw gate r_j, and the clipped activity a_j presented to the decoder.

residual RMS rises from 0.026 to $0.123\,\mathrm{N\,m}$ and the decoder moves $\|W_T - W_0\|_F = 0.37$; on Roll, Full3, Hard45, and Hardstep PES applies 4–12× the fixed residual RMS on both models. Where no gate opens (Roll, Yaw360; $r_j \leq 1.6\times10^{-3}$) PES lifts the driven membranes toward threshold ($v_j \leq 0.87$) and uses the graded tail. Decoder adaptation is therefore the principal lever: the CMG dynamics supply a bounded basis, latched or graded, on which PES shapes the correction. Furthermore, the leak term λ_W acts as classical robust σ-modification. At steady state, the tracking error it induces is analytically bounded by $\|\boldsymbol{\varepsilon}_{ss}\| \leq \frac{\lambda_W}{\eta\|\boldsymbol{a}_{ss}\|^2}\|\boldsymbol{s}_{ss}\|$. With the parameters in Table II, this guarantees an imperceptible steady-state bias of $\sim 10^{-3}$, permanently resolving theoretical concerns about leak-induced drift.

E. Neuron-Family Comparison

The Nengo benchmark covers two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG, LIF, ALIF, Izhikevich [21], and LIFRate, all sharing the bounded residual, saturation, and allocation interface; only the neuron dynamics and native-state-to-activity map differ. Fig. 7 shows two patterns. First, PES is the dominant lever in every family: fixed decoders rarely match their PES counterparts, even with more expressive dynamics, extending Sec. IV-D to every basis. Second, lower error is not free: some Izhikevich–PES cells reach the smallest error, but at an average normalized residual RMS near 0.37 versus 0.12 for CMG–PES and with larger decoder movement, so equal output caps do not imply equal realized effort. By Proposition 3, error contraction scales strictly by $(1 - \eta\|\boldsymbol{a}\|_2^2)$. Standard LIF/ALIF models suffer from sparse activity ($\|\boldsymbol{a}\|_2^2 \to 0$), which causes the multiplier to stall at 1, mathematically blunting learning. Conversely, the CMG unit's graded logistic tail maintains a resting activity (≈ 0.37), strictly bounding $\|\boldsymbol{a}\|_2^2 > 0$. This guarantees continuous error contraction and provides the formal mathematical reason why CMG breaks the effort-accuracy tradeoff. CMG–PES thus attains comparable tracking at markedly lower realized effort and more conservative decoder motion.

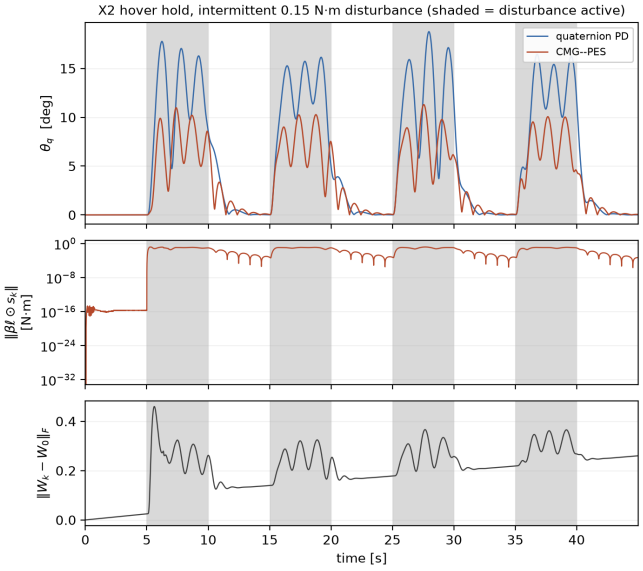


Fig. 6. X2 hover hold under four 0.15 N m disturbance bursts (shaded) separated by quiet intervals. Top: attitude error. Centre: residual magnitude, logarithmic. Bottom: decoder displacement from W_0. The residual is at numerical precision before the first burst. This means negligible initial correction, not zero computation or energy cost.

F. Disturbance Rejection and Retention

An equilibrium test applies $\boldsymbol{\tau}_{\mathrm{d}} = d_b \rho(t)[\sin(1.31t + .2), .8\sin(1.73t + 1.1), .6\sin(2.17t + 2)]^{\mathsf{T}}$ in body axes. After 5 s of quiet, ρ alternates 5-s on/off intervals with 0.25-s raised-cosine edge ramps. For X2, $d_b = .15$ N m is the first component's amplitude, not a vector-norm bound. The four-burst 45-s experiment (Fig. 6) cuts burst-phase mean error from $11.50°$ to $6.65°$ (42.2%) and post-burst quiet error from $1.21°$ to $.99°$ (18.0%). The CF2 rerun uses $d_b = .00015$ N m (a selected scaling, not an exact inertia-ratio match); its full-run mean falls from .141199 to .070259 rad (50.24%).

V. Discussion and Conclusion

A bounded residual of six continuous CMG units and eighteen PES-adapted decoder weights augments a fixed quaternion PD baseline without changing its gains or actuator interface, reducing mean quaternion-angle error by 17.3–60.6% over ten matched 45-s MuJoCo cases on two modeled platforms, within rotor saturation and with exact baseline recovery at zero authority; direct and Nengo realizations agree to 2.013×10^{-6} relative mean error.

Decoder adaptation is the dominant source of improvement in the tested regime: a fixed decoder gives at most 10.3%, and toggling χ is indistinguishable. Opened gates latch and saturate the readout, where a fixed paired decoder can cancel an axis, while sub-threshold units stay in a graded logistic tail; PES shapes the correction on either basis. Across all five neuron families PES is the consistent differentiator, and the dynamics mainly set an effort–accuracy tradeoff. The readout gain G, which sets the resting activity at 0.37, is a key sensitivity parameter.

Limitations: results are in simulation (Sec. IV-C bounds noise sensitivity; model mismatch and hardware effects are not addressed), the retention horizon is limited, and bounded commands, states, and weights are implementation properties, not global stability proofs.

Future work includes baselines with desired-rate feedforward and integral action, matched-effort comparisons against non-neural bounded bases, sweeps of G and χ, longer retention studies, held-out evaluations with frozen learned weights, full 6-DOF extension, and, most importantly, validation on physical aerial, ground, and other robotic platforms; the same bounded-residual wrapper extends to nonlinear and optimal-control baselines with hard actuator constraints.

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Fig. 7. Mechanism separation across the Nengo neuron-family benchmark, X2 (top) and CF2 (bottom). Left: mean q-error improvement over the PD baseline. Right: normalized residual effort, the RMS residual as a fraction of the per-axis limit. Rows are neuron family and learning mode, columns are missions; every cell uses the same bounded residual, saturation, and allocation interface. Two patterns hold on both platforms: PES adaptation, not the neuron model, separates the improving rows from the flat ones; and the families that match or exceed CMG–PES on error do so at three to twenty times its residual effort.

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